

Programme	B. Sc. Computer Science				
Course Title	Fundamentals of Computers and Computational Thinking				
Type of Course	Major				
Semester	I				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Fundamentals of electronic components 2. Basic mathematical operations				
Course Summary	This course provides a comprehensive overview of computing, covering historical milestones, hardware components, software systems, and computational thinking principles. Students will explore the evolution of computing systems, from early pioneers to modern processors and quantum units. The curriculum delves into hardware intricacies, software distinctions, and essential concepts in computer science, emphasizing problem-solving skills and algorithmic thinking. Practical aspects include hands-on experiences with hardware assembling, operating system installation, algorithm and flowchart visualization.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Develop a foundational knowledge of computing systems, encompassing their historical development, evolutionary milestones, and the notable contributions of key figures in the field.	U	F	Instructor-created exams / Quiz
CO2	Acquire familiarity with diverse hardware components constituting a computer system.	U	C	Practical Assignment / Observation of Practical Skills
CO3	Gain practical expertise by engaging in hands-on activities focused on the installation and configuration of diverse hardware components within a computer system.	Ap	P	Practical Assignment / Observation of Practical Skills

CO4	Explore the spectrum of software types, and actively participate in the partitioning, installation, and configuration of operating systems to cultivate a comprehensive understanding of software systems.	Ap	P	Practical Assignment / Observation of Practical Skills
CO5	Develop a foundational understanding of computer science as a discipline, examining problems through the lens of computational thinking and cultivating analytical skills to address challenges in the field.	An	C	Instructor-created exams / Quiz
CO6	Represent complex problems using algorithmic approaches and enhance problem-solving skills by visualizing solutions through the utilization of various software tools.	Ap	P	Practical Assignment / Observation of Practical Skills
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Fundamentals of Computers and Computational Thinking

Module	Unit	Content	Hrs	Marks
I	History and Evolution of Computing System		9	15
	1	Evolution of Computers – History, Generations	1	
	2	Overview of Computer System - Von Neumann Model, Number Systems(Binary, Hexa, Octal, Decimal)	2	
	3	Number Conversion and Digital Codes - Conversion from one number system to another, Digital Codes (Gray, Excess-3, BCD)	2	
	4	Pioneers and Contributors of Computing Systems - First Mechanical computer - Charles Babbage, Stored-Program Architecture - John von Neumann, Turing machine - Alan Turing, First General-Purpose Electronic Digital Computer -	2	
		John Mauchly and J. Presper Eckert, Artificial Intelligence- John McCarthy (Contributions only).		
	5	Computing Systems: Past to Present - Single Core, Dual-Core and Multi-Core Processors, Graphics Processing Unit (GPU), Accelerated Processing Unit, Quantum Processing Units (QPU) (Concept only).	2	

II	Hardware		11	20
	6	Electronic Components – Active Components - Diode, Transistor, Integrated Circuits (Definition, Symbol and Function).	1	
	7	Electronic Components - Passive Components – Resistors, Capacitors, Inductors (Definition, Symbol and Function).	1	
	8	Motherboard Components – CPU and Cooling Fan, RAM, Expansion Slots (PCIe), Input/Output Ports, Chipset (Concept only).	2	
	9	Motherboard Components – BIOS/UEFI Chip, SATA/NVMe Slots, Network Interface, Ports- Ethernet, VGA, HDMI, USB (Concept only).	3	
	10	Computer Components – SMPS, Motherboard, Storage Devices (HDD, SSD, NVMe)(Concept only).	2	
	11	Computer Components – RAM (DRAM, SRAM, DDR SDRAM), ROM, Cache (Concept only).	2	
III	Software		10	15
	12	Softwares - Application Software, System Software, Examples	1	
	13	Operating Systems – Need of OS, Types – Proprietary and Open Source, Hardware Software Compatibility, POST, Booting.	4	
	14	OS Installation – Bootable Media, UEFI / Legacy BIOS, Disk Partitioning, Dual Booting, Boot Manager – BOOTMGR, Grub, File Systems- FAT, NTFS, ext4.	4	
	15	Device Drivers – Need of Device Drivers, Driver Interactions (Basic concept only).	1	
IV	Computer Science and Computational Thinking		15	20
	16	Computer Science - Introduction, Role of Computer Science in the Modern Era	1	
	17	Problem Solving - Defining the Problem, Systematic Approach.	2	
	18	Computational Thinking – Problem Decomposition, Pattern Identification, Abstraction, Generalization.	2	
	19	Logical Thinking – Inductive and Deductive Reasoning, Logical Expressions.	2	
	20	Algorithmic Thinking – Intuition vs Precision, Defining algorithms.	2	
	21	Algorithm – Need of Algorithm, Qualities of a Good Algorithm, Examples.	3	
	22	Flowchart - Flowchart Symbols, Examples. Raptor.	3	
V	Lab Activities		30	30

	Some of the suggested lab activities are given below. 1. Identify, categorize and list out specifications of given electronic components. 2. Identify and list out specifications of given motherboard components. 3. Identify and Describe various ports and connectors on the motherboard. 4. Installation of various components on the motherboard (Processor, Fan, Heat Sink, RAM etc.) 5. Hands-on experience in assembling and disassembling a computer system (SMPS, Motherboard, Storage Device etc.). 6. Accessing and configuring the Basic Input/ Output System (BIOS) or Unified Extensible Firmware Interface (UEFI) settings. 7. Preparation of Bootable media with software like <i>Rufus</i>. 8. Check the hardware compatibility and Install operating system (single booting) on given computer. 9. Check the hardware compatibility and Install operating systems (dual booting – Windows and Linux) on a given computer. Develop algorithms and implement the solutions using <i>RAPTOR</i> flowchart execution tool for the following problems. 10. Read and print a number. 11. Read the price of three items and print the total bill amount. 12. Read the ages of two persons and print the elder one. 13. Read the number of units of electricity consumed and print the bill amount for various slabs. 14. Read a year and check whether it is a leap year. 15. Print first N numbers (using loop).		
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References:

1. Gary B. Shelly, Thomas J. Cashman, and Misty E. Vermaat. "Introduction to Computers", Cengage Learning, 2008.
2. Pradeep K. Sinha and Priti Sinha, Computer Fundamentals: Concepts, Systems & Applications. BPB Publications.
3. Kevin Wilson, Computer Hardware: The Illustrated Guide to Understanding Computer Hardware. Amazon Digital Services LLC – KDP, 2018.
4. John Hanna, OS Installation 101: A Step-by-Step Approach for Newbies.
5. David Riley and Kenny Hunt, Computational thinking for modern solver, Chapman & Hall/CRC, 2014
6. R.G. Dromey, How to solve it by Computer, PHI, 2008